PROGRAM 15 – DECISION TREE

AIM

To write a Python program to implement the Decision Tree algorithm using machine learning to predict whether a person will play cricket based on weather conditions.

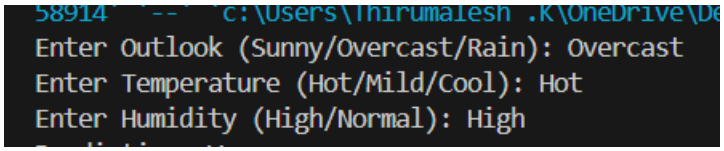
ALGORITHM

1. Start the program.
2. Import the DecisionTreeClassifier from the Scikit-learn library.
3. Create a dataset containing weather conditions and play decisions.
4. Convert categorical values into numerical values.
5. Separate input features and target output.
6. Train the Decision Tree classifier.
7. Read the weather conditions from the user.
8. Predict the result using the trained model.
9. Display whether the person will play or not.

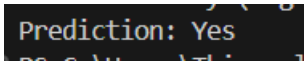
CODE

```
from sklearn.tree import DecisionTreeClassifier
outlook={"Sunny":0,"Overcast":1,"Rain":2}
temperature={"Hot":0,"Mild":1,"Cool":2}
humidity={"High":0,"Normal":1}
X=[[0,0,0],[0,1,0],[1,0,0],[2,1,0],[2,2,1],[1,2,1],[0,2,1],[2,0,1]]
y=["No","No","Yes","Yes","Yes","Yes","Yes","No"]
model=DecisionTreeClassifier(random_state=0)
model.fit(X,y)
o=input("Enter Outlook (Sunny/Overcast/Rain): ").capitalize()
t=input("Enter Temperature (Hot/Mild/Cool): ").capitalize()
h=input("Enter Humidity (High/Normal): ").capitalize()
if o in outlook and t in temperature and h in humidity:
    prediction=model.predict([[outlook[o],temperature[t],humidity[h]]])
    print("Prediction:",prediction[0])
else:
    print("Invalid input")
```

INPUT



OUTPUT



RESULT

Thus, the Decision Tree algorithm was successfully implemented using the Scikit-learn library to predict the result based on weather conditions.